MIL-I-6599C
8 December 1987
SUPERSEDING
MIL-I-6599B
24 MARCH 1967

MILITARY SPECIFICATION

INDICATOR, RATE OF FLOW, FUEL, TYPE A-19

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a type of single, remote indicating, fuel flow rate indicator, designated type A-19.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

Federal

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-I-7057	Indicator, Synchro, Aircraft, General Specification For

STANDARDS

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-2073-1	DOD Material Procedures For Development And Application Of Packaging Requirements

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-07-23T01:08Z
Check the source to verify that this is the current version before use.

MS33585	Pointer, Dial, Standard Design Of Aircraft Instrument
MS33639	Case, Instrument, Clamp-Mounted, Aircraft

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Soceity For Testing And Materials (ASTM)

ASTM D3951 Packaging Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organization which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements specified in MIL-I-7057 are applicable as requirements of this specification. Additional requirements shall be as specified herein. In the event the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.2 Design.

3.2.1 Calibration. Each indicator dial shall be graduated in accordance with figure 1. The calibration of fuel flow versus angular degrees shall be linear within $\pm 0^{\circ}$ 15' of arc, except at the changeover point.

3.2.1.1 Indicator. Unless otherwise specified, the indicator shall be designed to indicate, at a remote station, the rate of flow in pounds per hour (pph) of aviation fuel with a specific gravity of 0.770 at 15.6°C.

3.2.2 Dial.

Source: https://assist.dla.mil -- Downloaded: 2026-07-23T01:08Z
Check the source to verify that this is the current version before use.

3.2.2.1 Dial diameter. The dial shall have a minimum diameter of 1.750 inches, as measured at the outside ends of the outer scale graduations.

3.2.2.2 Fluorescent-luminescent markings. The following markings shall be finished in fluorescent-luminescent material. The dimension of the markings shall be as shown in table I.

TABLE I. Dimensions of dial markings.

Marking	Height or length inch ±0.016	Width of line or graduation inch ±0.005
Scale numerals 0 through 3	0.188	0.031
Scale numerals 4 through 12	0.125	.020
1,000-pound graduation 0 through 3	0.156	.031
1,000-pound graduation 4 through 12	0.125	.020
500-pound graduation 0 through 3	0.125	.020
100-pound graduation	0.094	.020
Lettering "PPH" "X 1000"	0.094	.016
Lettering "FUEL FLOW"	0.125	.020
Shaded portion of pointer	-----	----

3.2.2.3 Durable dull black. The markings "USAF" and "TYPE A-19" shall be permanently and legible marked on the dial in 0.047 inch letters. These markings and all other markings not otherwise specified shall be finished in durable dull black.

3.2.3 Pointer. The pointer shall conform to MS33585 except that the length of the pointer shall be such that the tip will overlap from one-third to two-thirds of the length of the shortest graduation.

3.2.4 Case. The case shall be hermetically sealed and shall conform to MS33639. The dimension "L" shall be a maximum of 3 inches.

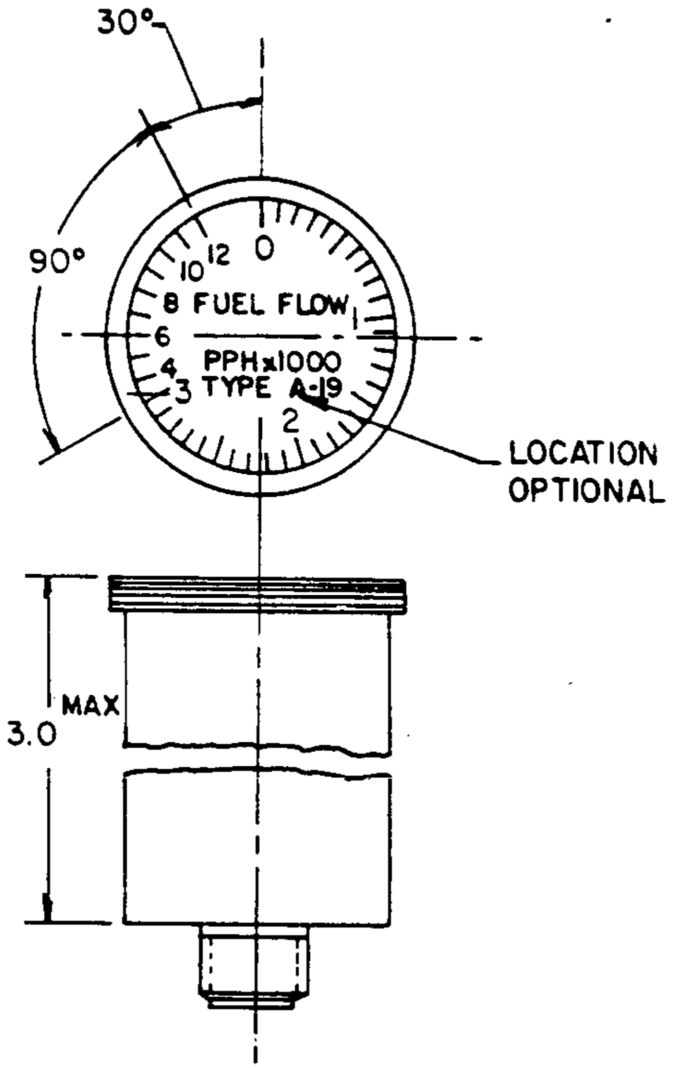
3.3 Weight. The weight of the complete indicator shall not exceed 0.60 pound.

3.4 Identification of product.

3.4.1 Nameplate. The nameplate shall be marked with the following information:

INDICATOR, RATE OF FLOW, FUEL,
Type A-19
26V 400 Cycles
Specification MIL-I-6599

Source: https://assist.dla.mil -- Downloaded: 2026-07-23T01:08Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES:

ANGLES ± 0° 15'

FIGURE 1. Dial.

Source: https://assist.dla.mil -- Downloaded: 2026-07-23T01:08Z
Check the source to verify that this is the current version before use.

Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's name or trademark
US Property

3.5 Recycled and reclaimed materials. Recycled and reclaimed materials shall be used to the maximum extent possible without jeopardizing the end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective matetrial, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 General. The smapling, inspection, and test procedures shall be in accordance with MIL-I-7057.

4.3 Performance. The performance of the indicator shall be checked against tables II and III of this specification.

4.4 Inspection of packaging. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 5 of this specification.

4.5 Inspection conditions. Unless otherwise specified all inspections shall be performed in accordance with the test conditions specified in this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2)

5.1.1 Level A.

Source: https://assist.dla.mil -- Downloaded: 2026-07-23T01:08Z
Check the source to verify that this is the current version before use.

TABLE II. Indicator fuel flow rate tolerances.

Flow rate pph	Tolerance pph±		
	Room temp.	Low temp.	High temp.
100	10	15	10
200	10	15	10
300	10	15	10
600	10	15	10
900	10	15	10
1,200	10	15	10
1,500	10	15	10
1,800	10	15	10
2,100	10	15	10
2,400	10	15	10
2,700	10	15	10
3,000	10	15	10
4,000	100	150	100
5,000	100	150	100
6,000	100	150	100
7,000	100	150	100
8,000	100	150	100
9,000	100	150	100
10,000	100	150	100
11,000	100	150	100
12,000	100	150	100

TABLE III. Indicator test tolerances.

Test	Tolerance pph ±	
	0-3,000 Scale	3,000-12,000 Scale
Position error at 2,500 pph	10	--
Friction		
Room temperature	20	200
High temperature	20	200
Low temperature	40	400
Vibration		
Pointer oscillation	20	200
Pointer variation	20	200

Source: https://assist.dla.mil -- Downloaded: 2026-07-23T01:08Z
Check the source to verify that this is the current version before use.

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity each indicator shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2)

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment form the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129 (see 3.4.1).

Source: https://assist.dla.mil -- Downloaded: 2026-07-23T01:08Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The type A-19 indicator covered by this specification is intended for use to indicate the rate of flow of fuel in pounds per hour consummed by an aircraft turbojet engine.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of preservation, packaging and packing.

c. Unit quantities.

6. Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL No.) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 and information pertaining to qualification of products may be obtained from that activity.

6.4 Subject term (key word) listing.

Fuel
Indicator
Rate of Flow
Remote
Type A-19
0-12,000 pph

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
AIR FORCE -99
NAVY -AS

Preparing activity:
AIR FORCE -71
Project number:
6620-0436

Source: https://assist.dla.mil -- Downloaded: 2026-07-23T01:08Z
Check the source to verify that this is the current version before use.